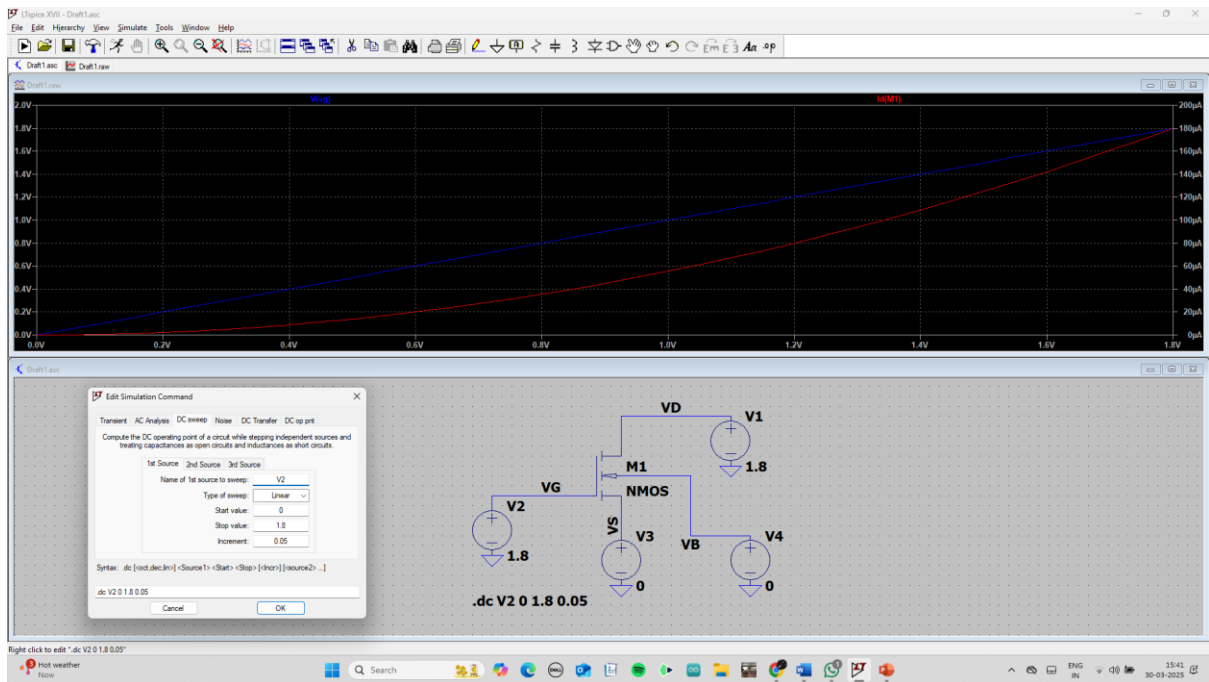
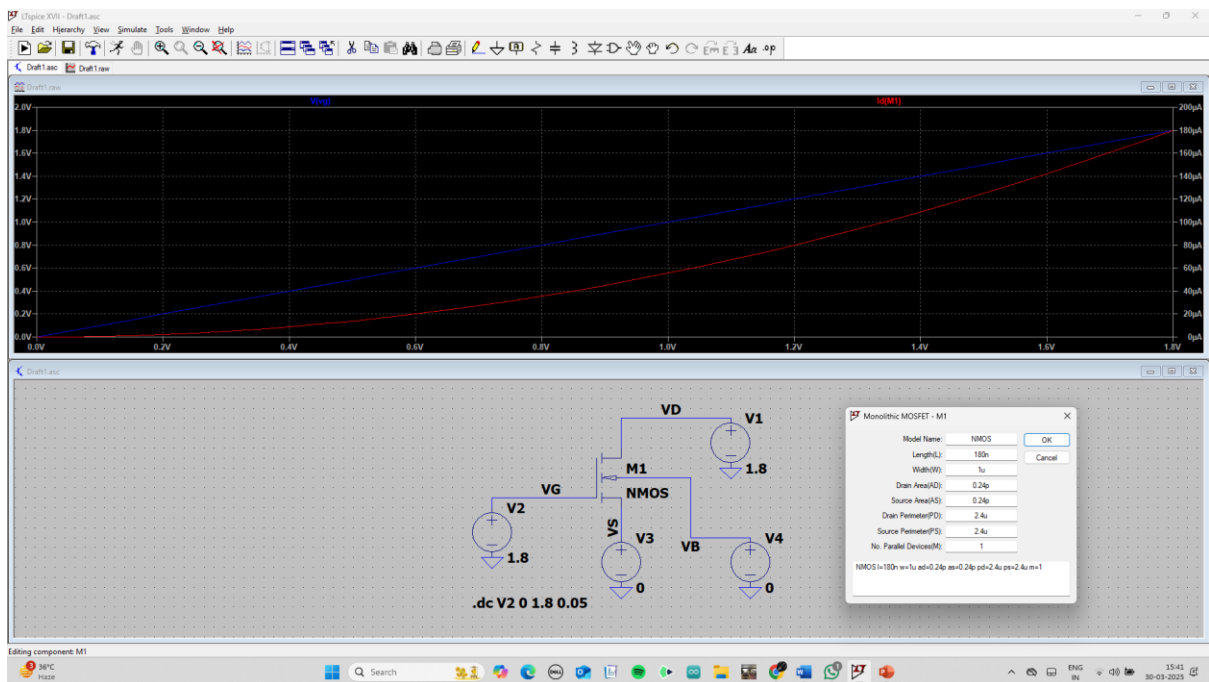
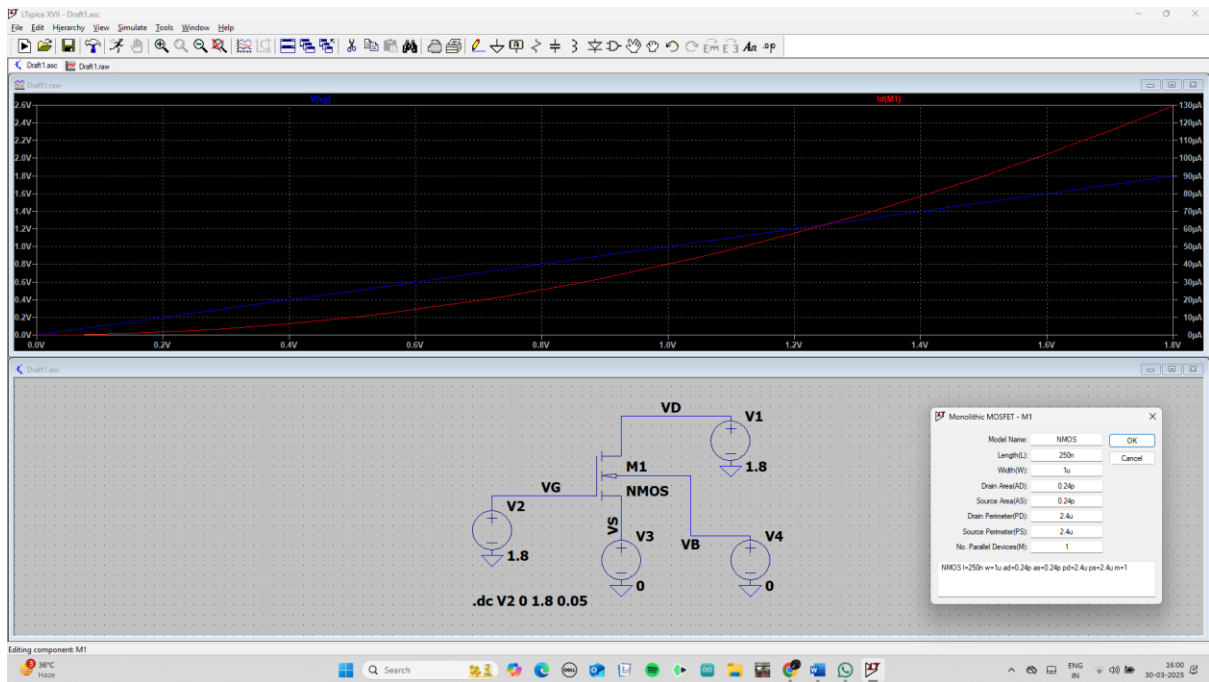




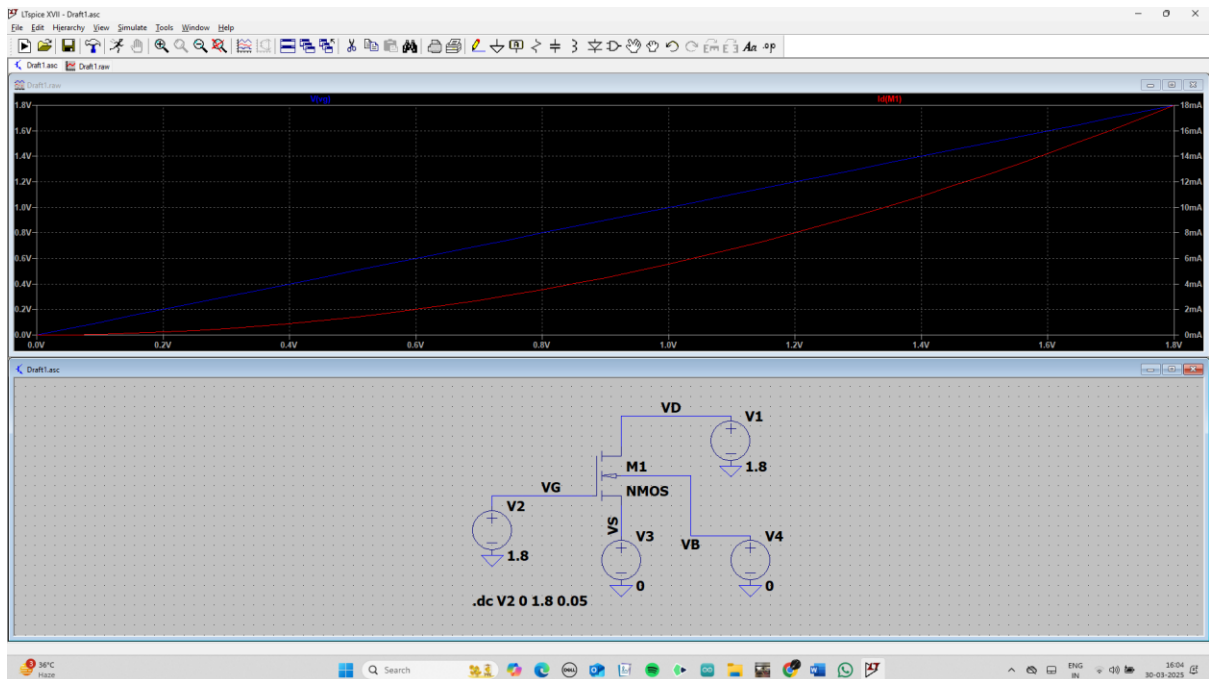
ID vs VGS

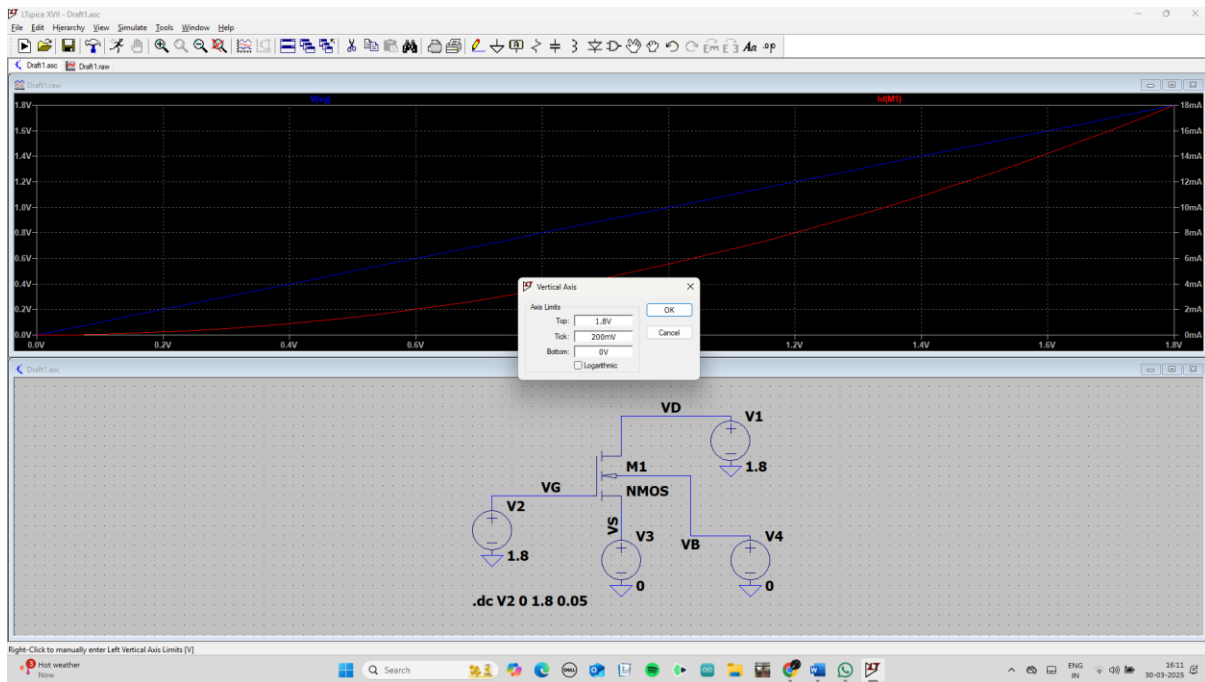


ID vs VGS with the following paramters of $L = 180\text{nm} = 0.18\mu\text{m}$ & $W = 1\mu\text{m}$, ID is $180\mu\text{A}$.



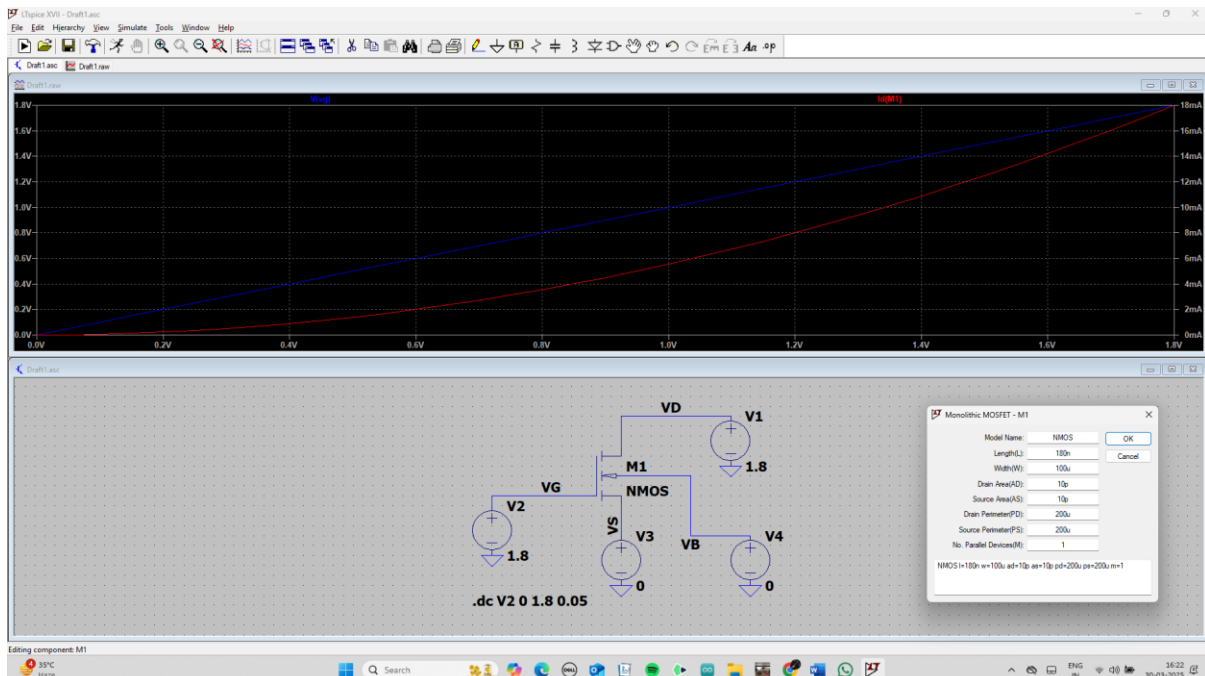
Now , Let's increase the W from 1um to 100um keeping L = 180nm as constant and let's see the change , theoretically ID should increase proportionally .





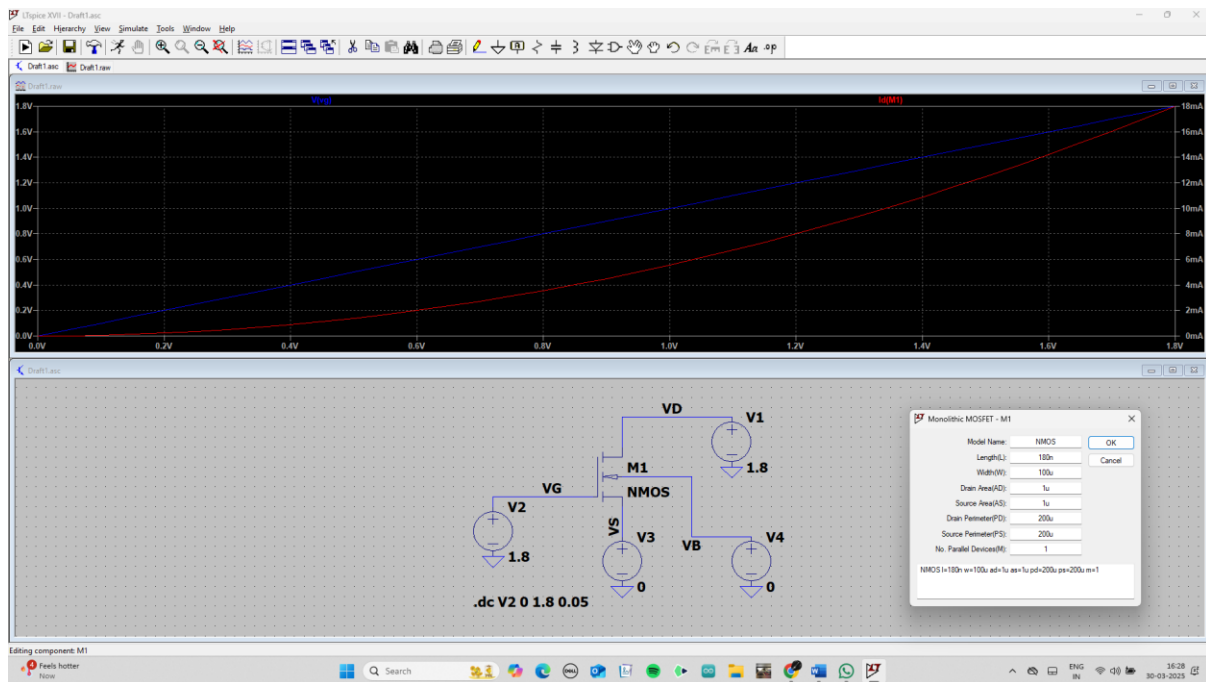
If we will Right click on the y-axis , and check the logarithmic scale , we will get **logID vs VGS** graph , Henceforth we can analyse the MOSFET's operation in SUB-THRESHOLD REGION !! that thing we will do it later on actually

Now let's see with the same specifications of MOSFET , and keeping everything constant Let's change the S & D perimeter and Area and see what happens let's evaluate . . .

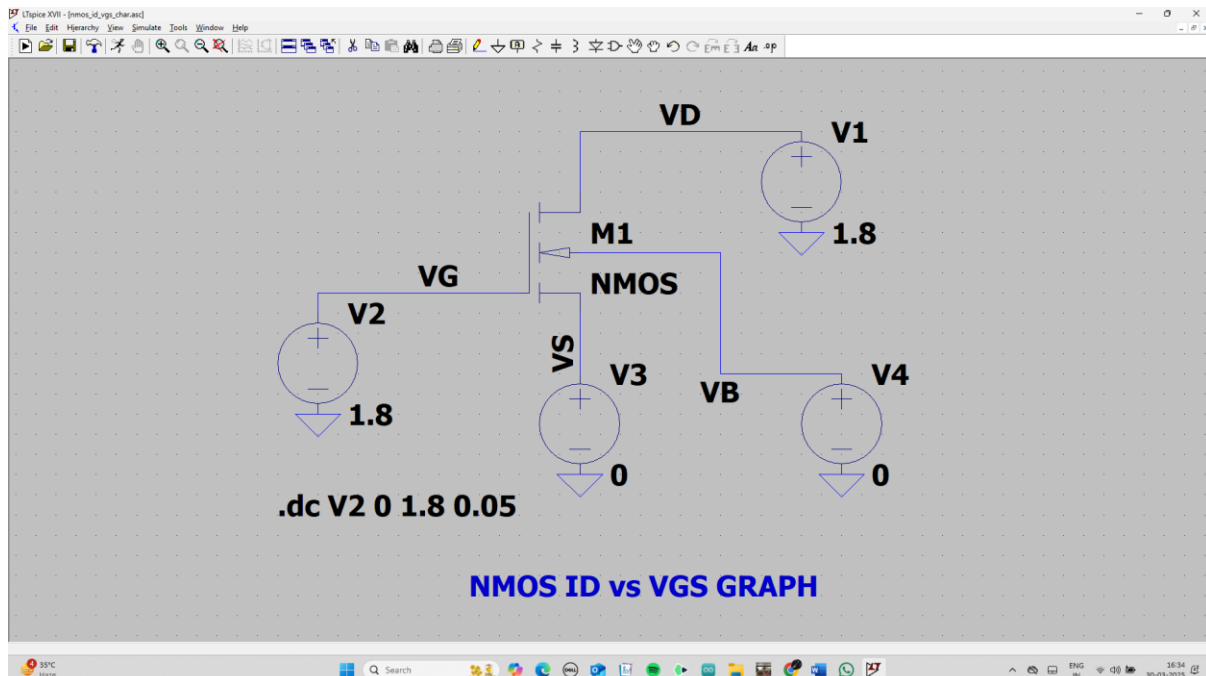


So , NO !! “ *keeping the W/L constant i.e., 100u/0.18um and increasing the AD , AS , PD , PS doesn't modify (increase or decrease) the ID in DC ANALYSIS !* ”

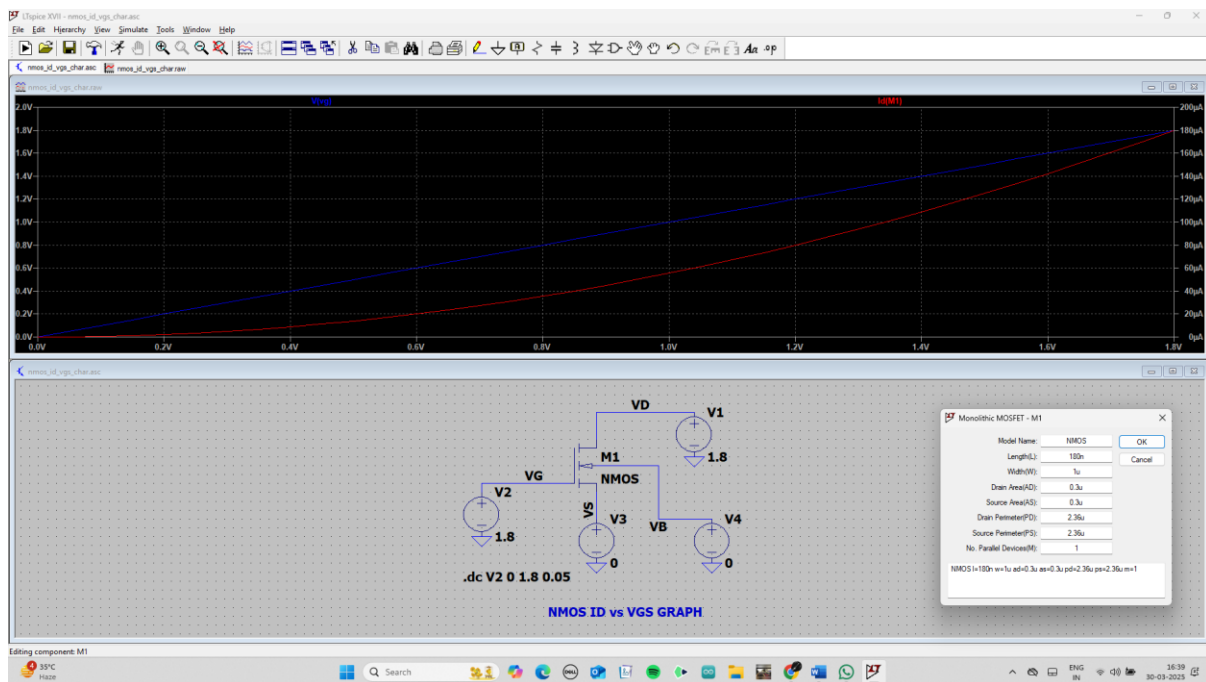
So , No matter how much you increase or decrease the AS , AD , PS , PD , ID will not change at all in DC ANALYSIS , But in AC ANALYSIS & TRANSIENT ANALYSIS , ID will change due to parasitic resistances and junction capacitances !!



In the above , simulation I have increased the AS , AD , PS , PD but ID remained the same i.e., 18mA ! But remember this is DC ANALYSIS !!



So , Throughout the analysis we will take an optimum W/L ratio with PS , PD , AS , AD as this :



We will get $I_D = 180\mu A$ that is standard !

Actually it's V_T is very very small , that's why I_D is getting much more at $V_G = 1.8V$

So , Let's do the AC analysis or TRANSIENT Analysis . . .

